

Trainings ▾

Preparatory Steps

WARNING

Do not remove the pressure cap from a hot engine. Wait until the coolant temperature is below 50°C [120°F] before removing the pressure cap. Heated coolant spray or steam can cause personal injury.

WARNING

Coolant is toxic. Keep away from children and pets. If not reused, dispose of in accordance with local environmental regulations.

WARNING

This component or assembly weighs greater than 23 kg [50 lb]. To prevent serious personal injury, be sure to have assistance or use a hoist or use appropriate lifting equipment to lift this component or assembly.

- Drain the coolant. Refer to Procedure 008-018 in Section 8.
(/qs3/pubsys2/xml/en/procedures/28/28-008-018-tr.html)
- Remove the exhaust manifold. Refer to Procedure 011-007 in Section 7.
(/qs3/pubsys2/xml/en/procedures/28/28-011-007-tr.html)
- Remove the exhaust manifold, wet water connections, is equipped. Refer to Procedure 011-016 in Section 11. (/qs3/pubsys2/xml/en/procedures/28/28-011-016-tr.html)
- Remove the lubricating oil cooler covers. Refer to Procedure 007-045 in Section 7.
(/qs3/pubsys2/xml/en/procedures/28/28-007-045-tr.html)

Remove

CAUTION

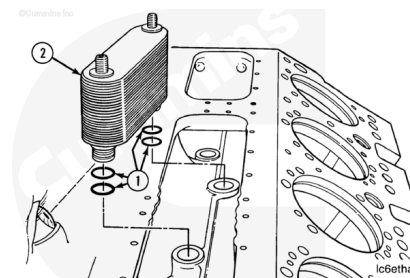
Do not damage the elements by prying with tools that have sharp or pointed edges or corners. The sharp tools can penetrate the element and cause a leak.

Note : A portion of the cylinder block is **not** shown for clarity.

Remove the three oil cooler elements (2) on the K38 and QSK 38 engines.

Remove the four oil cooler elements (2) on the K50 and QSK 50 engines.

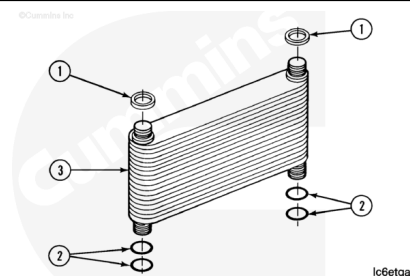
Remove and discard the o-rings (1) from each oil cooler element (2).



Clean and Inspect for Reuse

Note : Cummins, Inc. does **not** recommend any form of cleaning of oil cooler elements (internal or external). This includes cleaning with solvent, flushing with oil, steam cleaning and ultrasonic cleaning.

Remove the gaskets (1) from the oil cooler element (3).

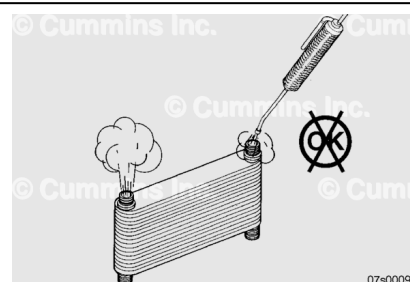


If the engine has experienced a malfunction that allows debris to pass through the lubrication system to the full-flow oil filters and oil passages resulting in the lubricating oil cooler become contaminated, the oil cooler elements **must** be replaced.

Inspect the oil cooler elements externally for scale build-up, corrosion, embedded debris, and/or any damage to the elements. If there is any evidence of any of these conditions, the oil elements **must** be replaced.

Note : Contamination can be defined by metal, rubber, or other solid particles present in the lubricating oil filter and/or by debris damage to the bearings, or debris damage of any other components such as they do **not** meet reuse guidelines.

Remove gasket material from the oil cooler cover, oil cooler element, and the engine block mating surface.



⚠ WARNING ⚠

When using solvents, acids, or alkaline materials for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing to reduce the possibility of personal injury.



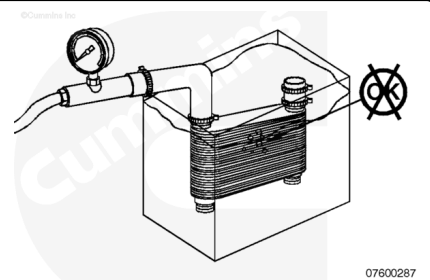
Clean the covers with solvent.

Inspect the covers for corrosion.

Pressure Test

Plug the oil cooler element, as illustrated, and connect to a source of compressed air.

Pressure-test each oil cooler element to check for leaks.



Measurements

	kpa	psi
Air Pressure:	517	75

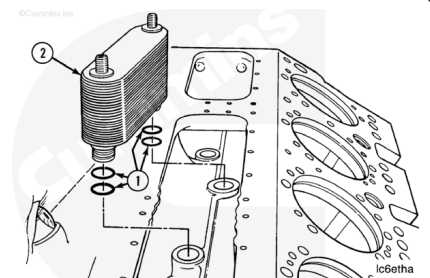
Note : Heating the water in the tank to 50°C [122°F] will improve the test results.

If the oil cooler element leaks, it **must** be replaced. Do **not** attempt to repair an oil cooler element.

Install

⚠ CAUTION ⚠

Check to be sure all of the shipping plugs have been removed from the elements or the engine will be damaged due to lack of lubrication.



Note : The K38 engine contains three oil cooler elements. The K50 contains four oil cooler elements.

Note : A portion of the cylinder block is not shown in the illustration for clarity.

Install the o-rings (1) in the grooves in each end of each cooler element (2).

Use vegetable oil to lubricate the o-rings and the bore in the block.

Install the element in the cylinder block. Use your hand. Push the element into the bore with hand pressure until all o-rings are in the bore completely.

Finishing Steps

WARNING

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WARNING

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- Install the lubricating oil cooler covers. Refer to Procedure 007-045 in Section 7. (</qs3/pubsys2/xml/en/procedures/28/28-007-045-tr.html>)
- Install the exhaust manifold. Refer to Procedure 011-007 in Section 7. (</qs3/pubsys2/xml/en/procedures/28/28-011-007-tr.html>)
- Install the exhaust manifold, wet water connections, if removed. Refer to Procedure 011-016 in Section 11. (</qs3/pubsys2/xml/en/procedures/28/28-011-016-tr.html>)
- Fill the engine with coolant. Refer to Procedure 008-018 in Section 8. (</qs3/pubsys2/xml/en/procedures/28/28-008-018-tr.html>)
- Operate the engine and check for leaks.